



Oral Surgery

Local anesthesia in the mandible

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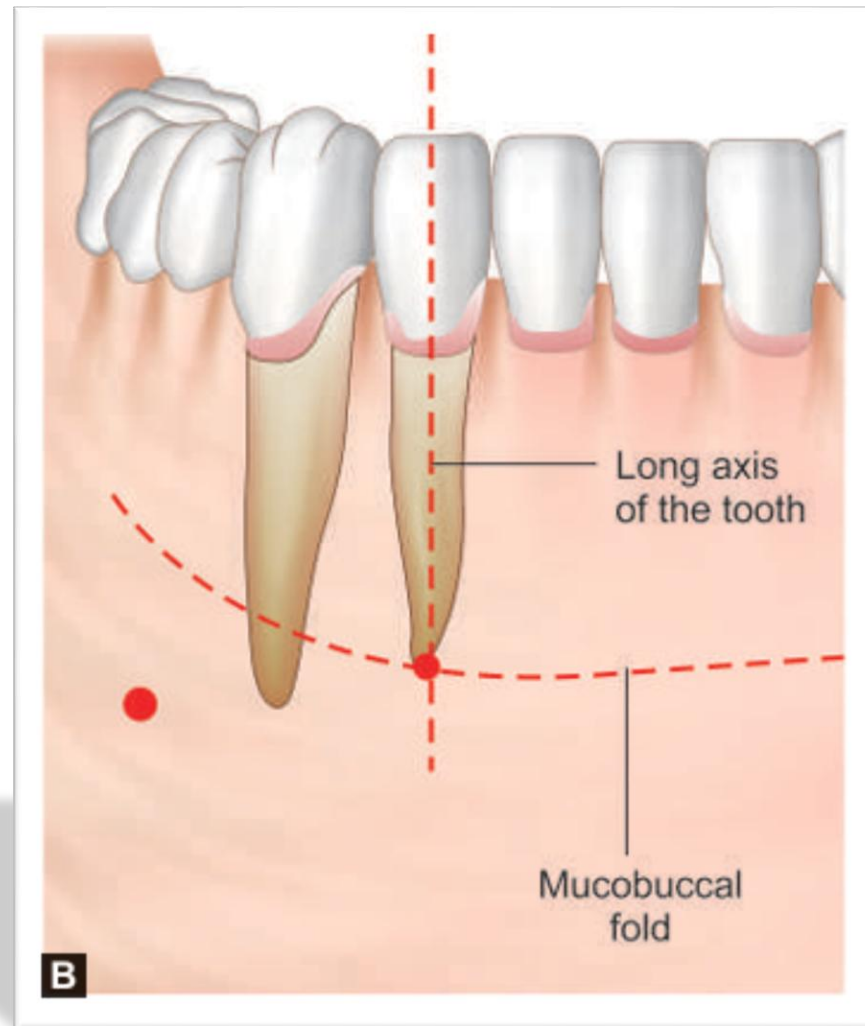
Anesthesia of the lower anterior teeth

The lower anterior teeth are supplied by the terminal branch of the inferior alveolar nerve (incisive nerve). Fortunately, the labial alveolar plate in this region is thinner and more porous than elsewhere in the mandible also there is considerable overlap in the midline, so this anastomosis makes an inferior alveolar block ineffective in the incisor region.

Because of this fact, the infiltration technique preferred be used in this area.

Anesthesia of the lower anterior teeth

This is achieved by deposition of about 1 ml of the solution in the labial sulcus adjacent to the target tooth to anesthetize the pulp and labial supporting tissues, which is enough to produce pulpal anesthesia of the teeth concerned, **but if surgery is to be done** up to 0.5 ml must be injected into the lingual sulcus.





Anesthesia of the lower premolars and molars

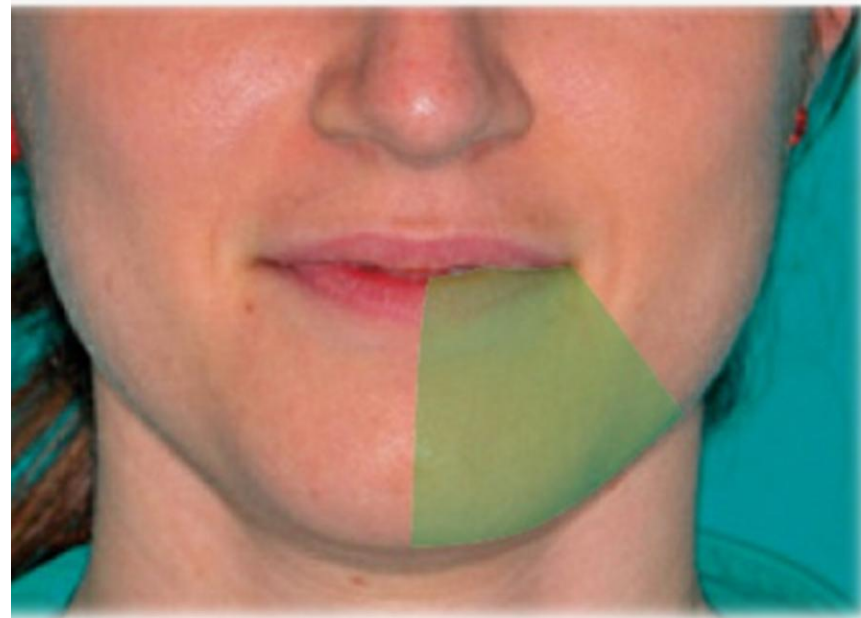
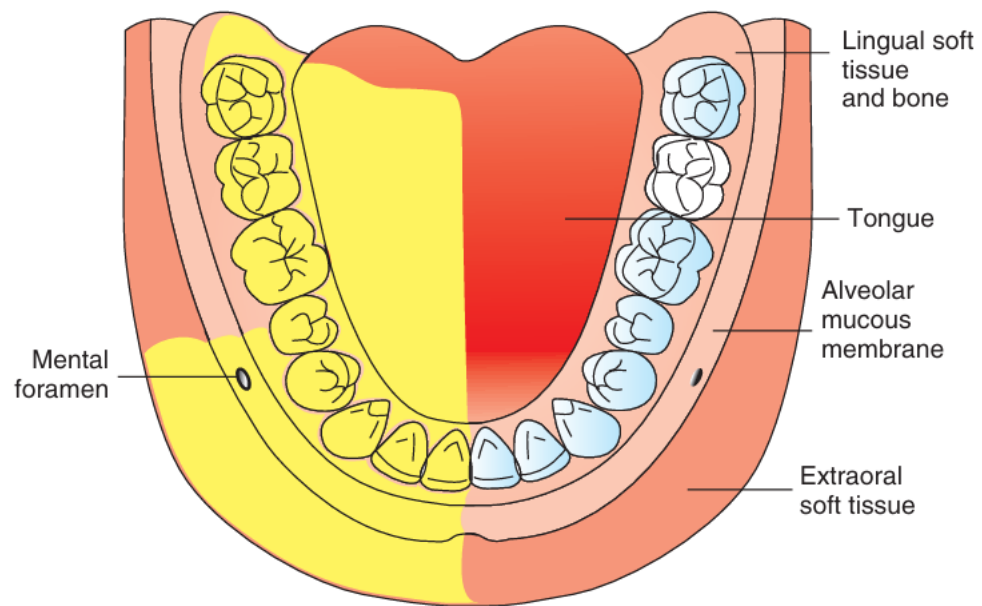
To extract the lower premolar and molars we should anesthetize the **inferior alveolar nerve** (supply the pulp of these teeth with their periodontium and the bony socket), **the lingual nerve** (supply the lingual soft tissue adjacent to them) and the **long buccal nerve** (supply the buccal soft tissue).

Anesthesia of the lower premolars and molars

In children, multiple vascular canals perforate the thin labial/buccal alveolar plate. For this reason, **infiltration techniques** are highly effective in producing anesthesia of upper and lower deciduous teeth. Care should be taken to avoid the misjudging the length of the roots and inserting the needle too deeply into the tissue.

Inferior alveolar nerve block

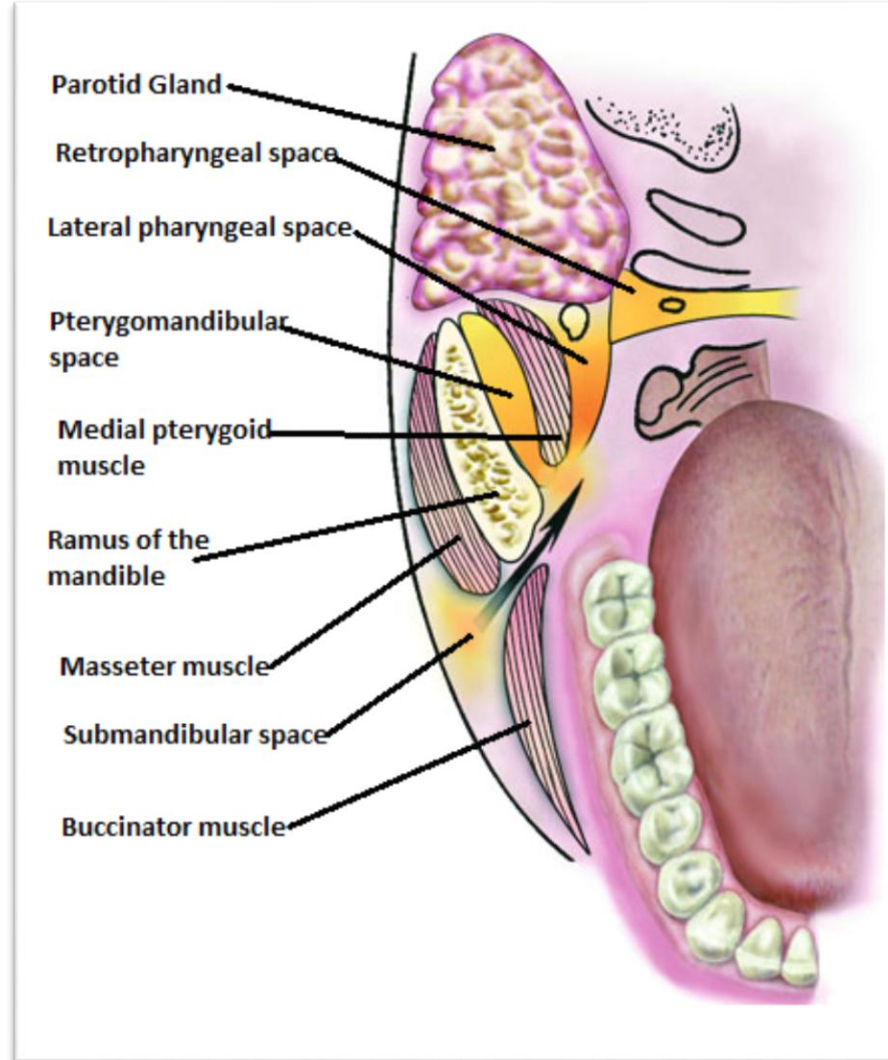
- The IAN, along with its terminal branches (incisive and mental nerves) are anesthetized by this technique (anesthesia of the pulp of all mandibular teeth in the same side of injection, the skin of the chin and the skin and mucosa of the lower lip). This is achieved by the deposition of the solution around the mandibular foramen in the pterygomandibular space.

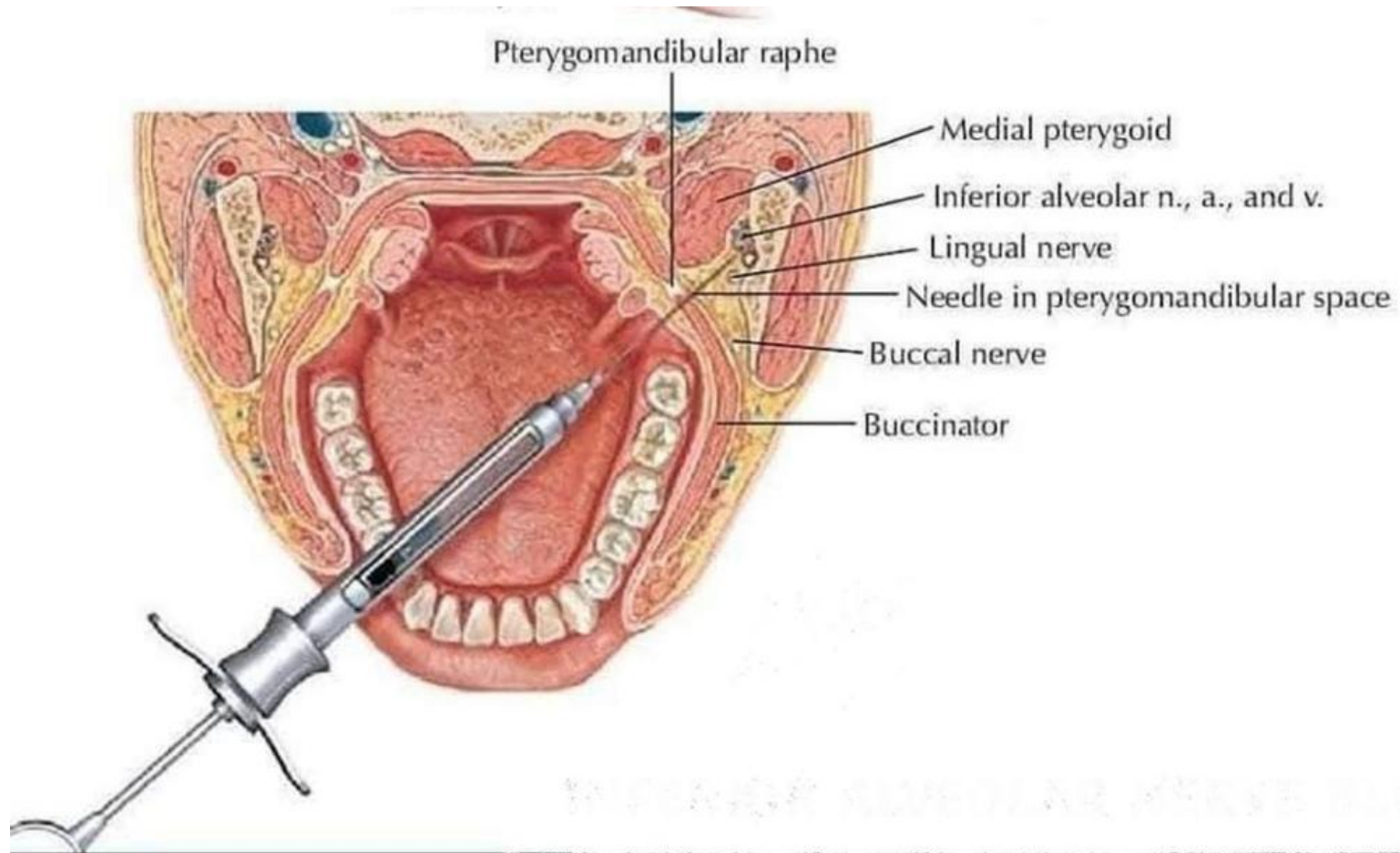


Figures illustrates the area anesthetized by using IAN block .

Boundaries of the pterygomandibular space

This space is bounded anteriorly by the pterygomandibular raphe and the fibers of superior constrictor and buccinator muscles that are inserted into it. The posterior boundary is formed by the parotid gland. The ramus of the mandible forms the lateral wall while the medial pterygoid muscle forms the medial wall. The lateral pterygoid muscle forms the roof of the space.





The inferior alveolar nerve block technique

The success of this technique is entirely dependent upon the accurate deposition of the solution; this technique will anesthetize the inferior alveolar nerve and its terminal branches.

The inferior alveolar nerve block technique

Technique: the patient is seated in the chair and the headrest adjusted so that his mandibular occlusal plane is horizontal when the mouth is open. The dentist should stand in front of the patient for the right side nerve block and behind the patient for the left.

The inferior alveolar nerve block technique

The mouth should be opened widely, the thumb of the free hand should rest on the **coronoid notch** and the **pterygomandibular raphe** should then be identified (thick soft tissue band originated from the pterygoid hamulus and inserted on the lingual aspect of the mandible in the third molar region).

The inferior alveolar nerve block technique

The syringe should be introduced from the lower premolar teeth of the other side parallel to and 0.5-1 cm above the lower occlusal plane such that the needle penetrates **laterally** to the pterygomandibular raphe.

The inferior alveolar nerve block technique

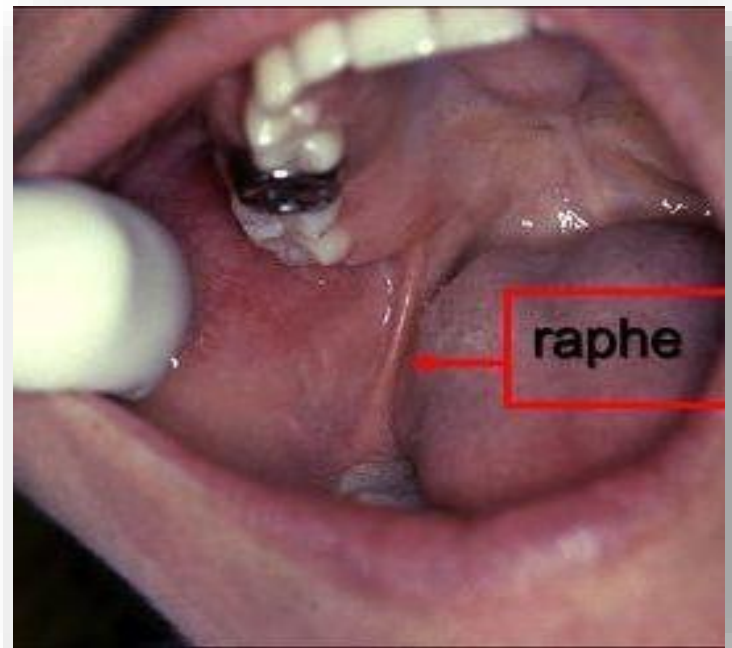
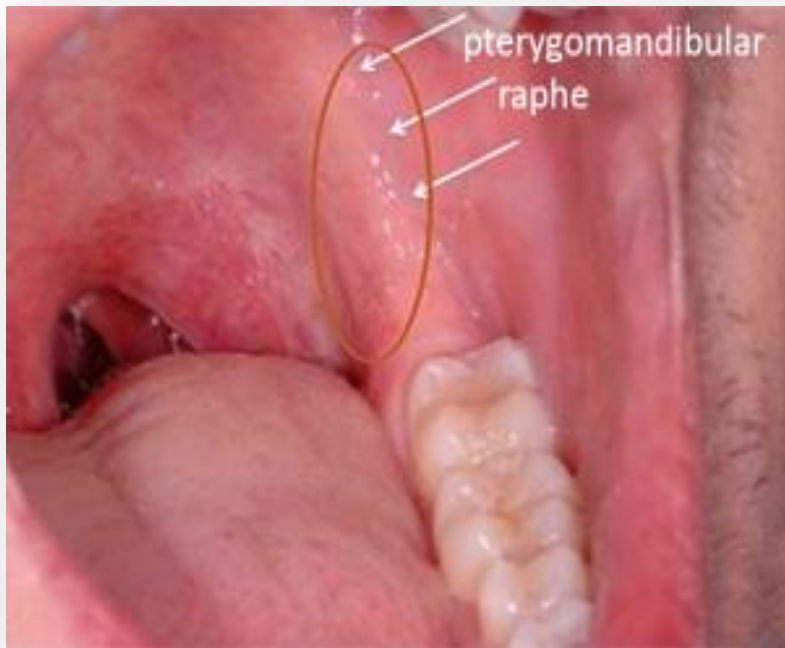
The height is at the level of the **coronoid notch**, the most concave region on the anterior border of the mandibular ramus. An alternate guideline for determining the correct height of entry for the IANB includes inserting the needle approximately **1 cm above the lower** occlusal plane when the mouth is fully open. Other landmarks include locating a level **midway between** the upper and lower dental arches when the mouth is wide open.

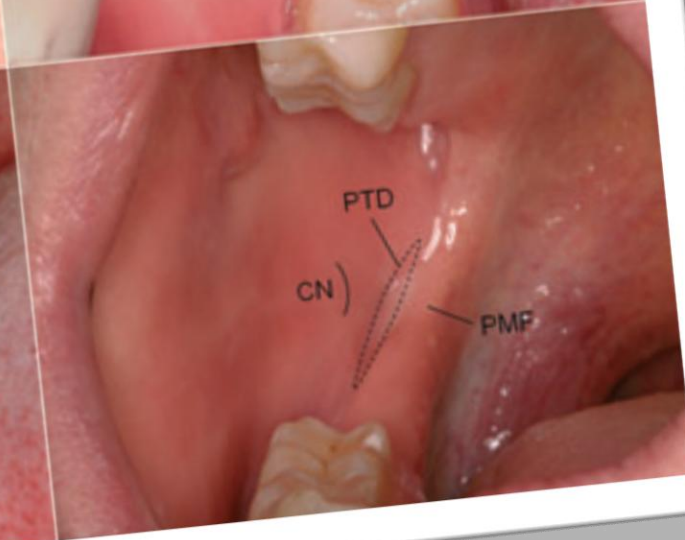
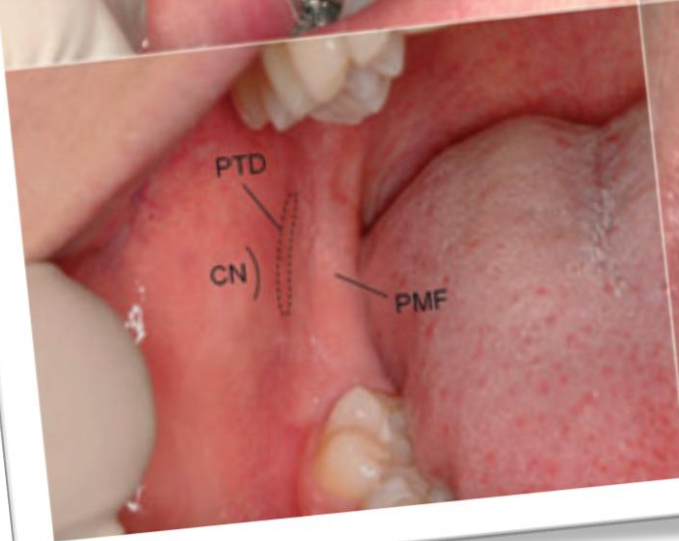
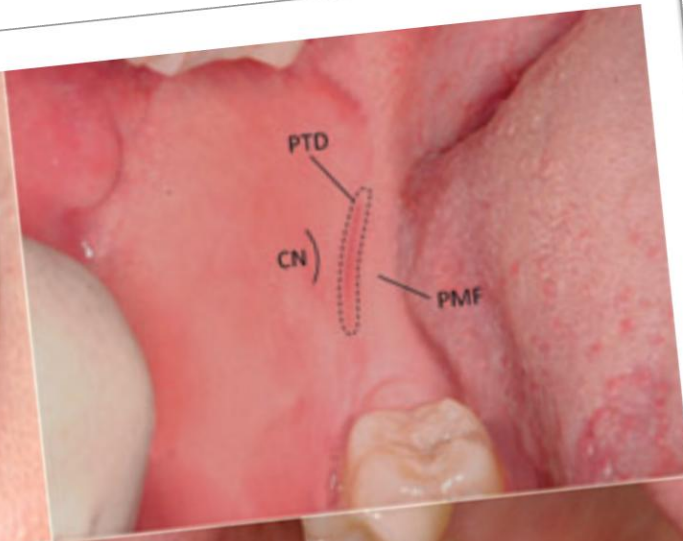
The inferior alveolar nerve block technique

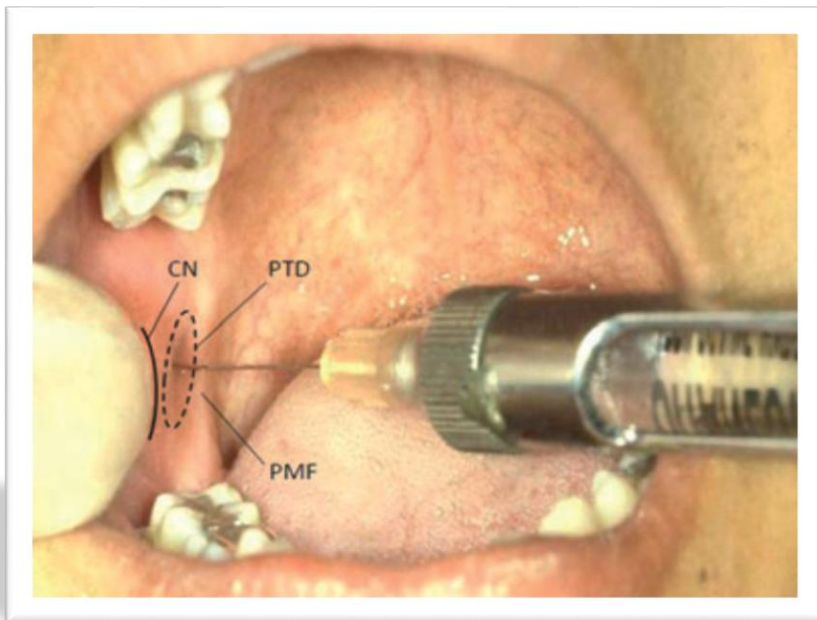
The long dental needle should be advanced **about 2-2.5 cm** until the bone is touched lightly, the needle should then be withdrawn a millimeter or two, the local anesthetic should then be deposited slowly using (1,5-2) ml of solution.

The inferior alveolar nerve block technique

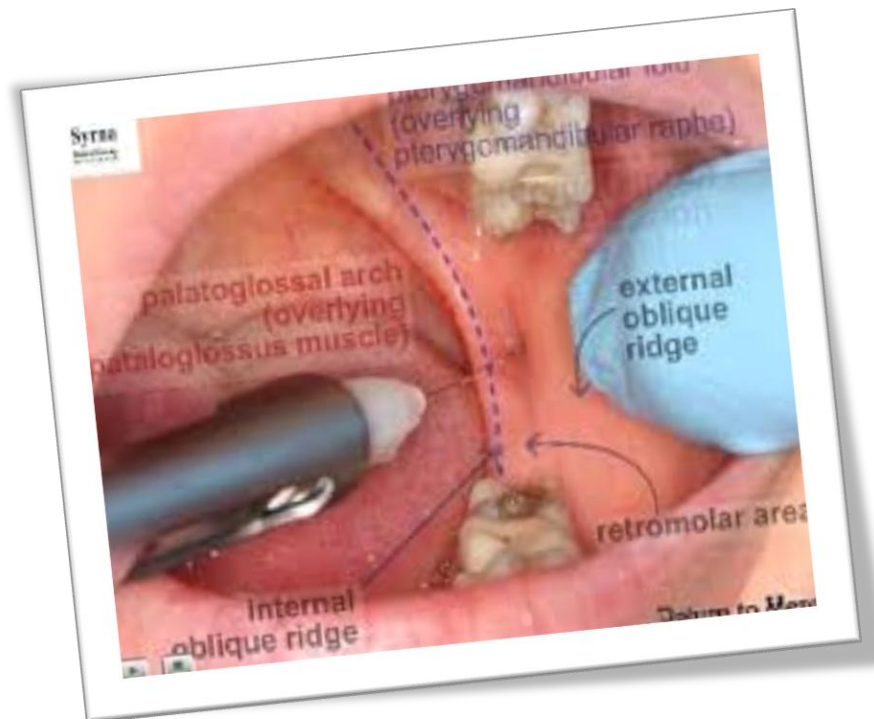
The **lingual nerve**, which lies anteromedial to the inferior alveolar nerve is anesthetized by slowly **withdrawing** the syringe until approximately half the needle's length remains within the tissue, deposit few drops of the solution.





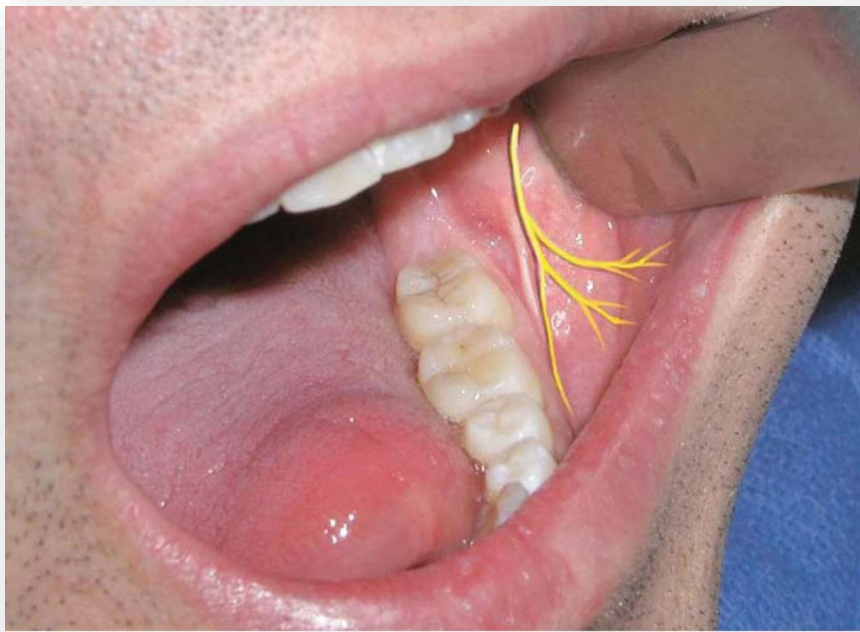


- (CN = coronoid notch; PTD = pterygotemporal depression; PMF = pterygomandibular fold.)



Long buccal nerve block

The long buccal nerve provides sensory innervations to the buccal soft tissue adjacent to the mandibular molar teeth, vestibular mucosa and mucosa of the retromolar fossa. A long buccal nerve block is achieved by means of a sub-mucous injection in which the solution is deposited (few drops) just posterior and buccal to the last molar tooth in the arch . Infiltration technique for the long buccal nerve is achieved by deposition of the solution in the muco-buccal fold adjacent to the target tooth.



A



B



C

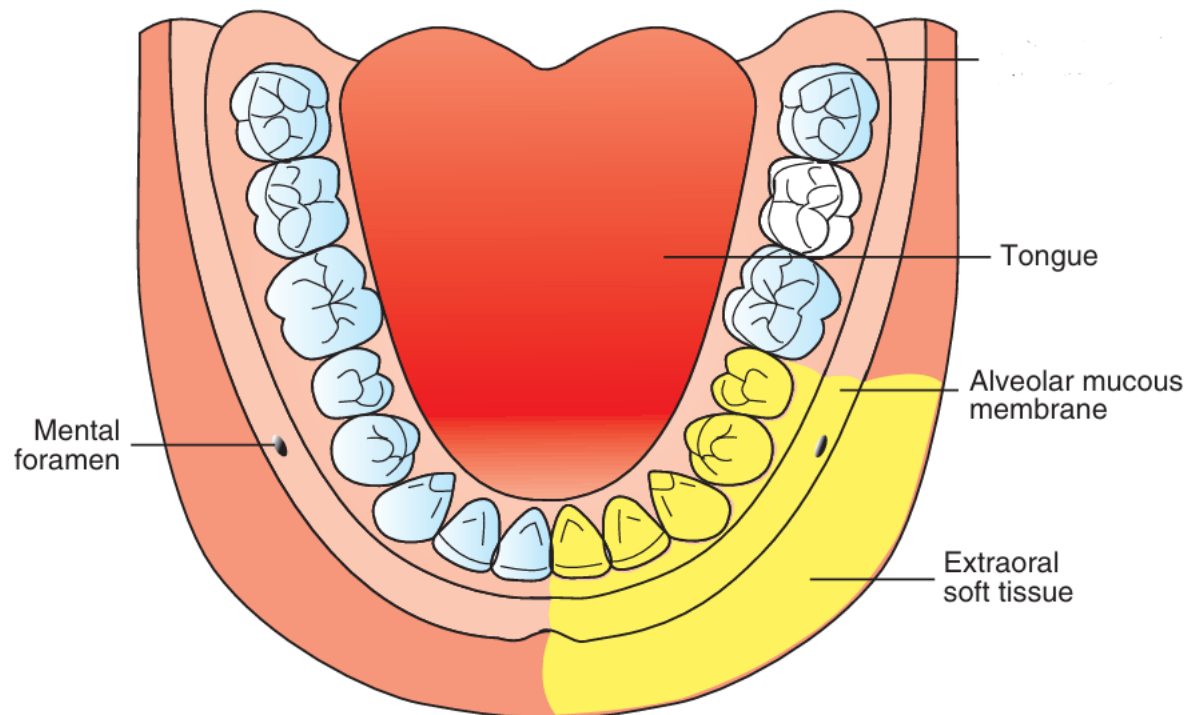
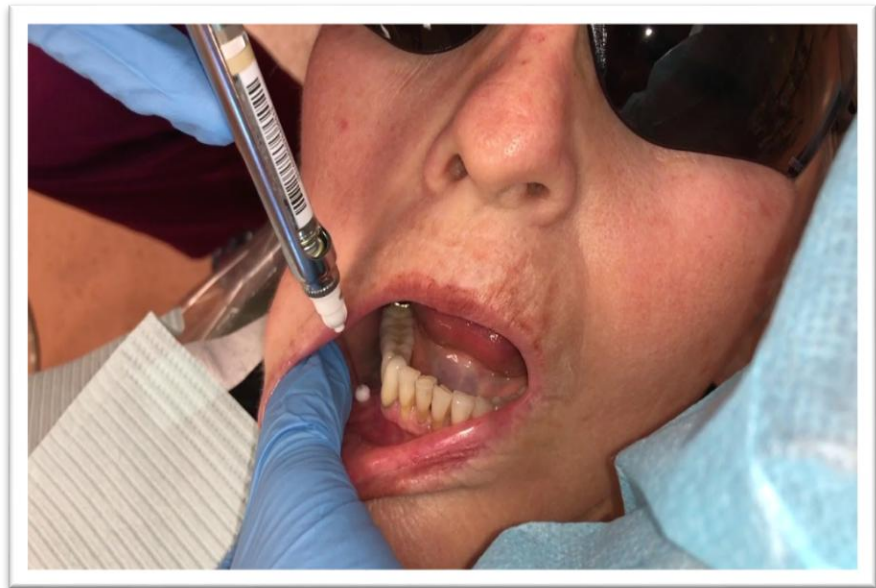
The mental nerve block

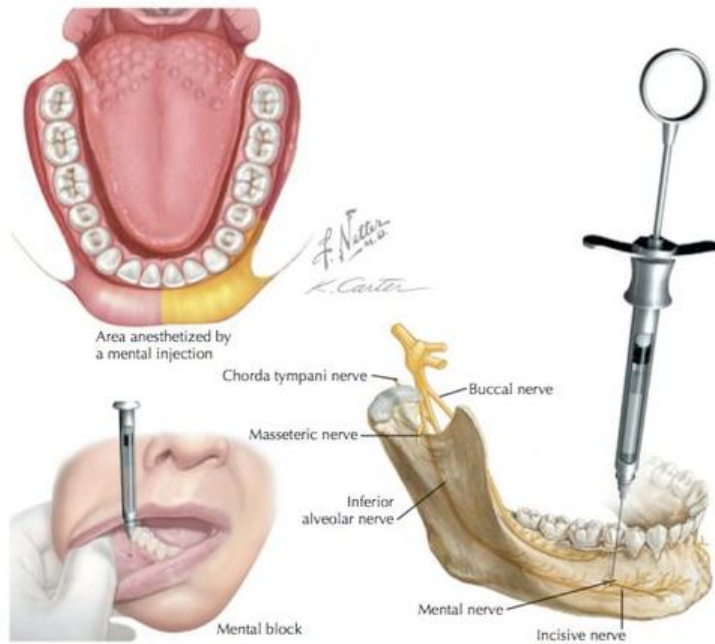
This block will anesthetize the pulp and periodontal membrane of the lower incisors, canine, first premolar and sometimes the second premolar (incisive nerve), labial mucous membrane anterior to the mental foramen and the skin of the chin and the skin and mucosa of the lower lip (mental nerve).

The mental nerve block

In dentate patient, the mental foramen lies below and between apices of the lower premolar teeth, approximately halfway between the cervical margin of the teeth and the lower border of the mandible. The solution is deposited at or near the foramen (about 1 ml).

In edentulous patients, the foramen may lie near the crest of the ridge because of bone resorption.





Mandibular nerve block

The entire mandibular branch of the trigeminal nerve is anesthetized.

The mandibular nerve could be blocked by

1. Intra-oral approach as in Gow-Gates mandibular nerve block and Akinosi (closed mouth) mandibular nerve block.
2. Extra-oral mandibular nerve block.

Mandibular nerve block

Gow-Gates mandibular nerve block

This technique is indicated when the conventional inferior alveolar nerve block is unsuccessful.

Akinosi (closed mouth) mandibular nerve block

It is indicated when there is limited mandibular opening and inability to visualize the landmarks for the inferior alveolar nerve block.

Mandibular nerve block

Extra-oral mandibular nerve block

This technique is indicated when there is an acute infection at the site of injection for the subdivisions of the mandibular nerve, when there is need to anesthetize the entire mandibular nerve and its branches with one single penetration and minimum amount of local anesthetic solution, and for diagnostic and therapeutic purposes as in trigeminal neuralgia.

Speed of injection

Rapid injection of the solution may cause **tissue distension and discomfort or actual cellular damage and pain** when sensation returns, also the risk of toxic reactions is greater if a quantity of anesthetic solution is injected rapidly, a slow rate of injection is thus essential.

In general, the ideal rate of injection is **1 mL/min.**

Testing for anesthesia

Altered sensation in itself does not guarantee that anesthesia has been achieved, prior to extraction under local anesthesia a dental probe may be pushed into the gingival crevice on both the labio-buccal and the lingual surface of the tooth.

The patient should be told that pressure might be felt and asked to state if sharpness is noted, the presence of sharpness is an indication of a further injection.

A decorative card with a light gray wooden plank background. The words "Thank You" are written in a large, black, cursive script in the center. The card is framed by several large, vibrant flowers in shades of orange, pink, and yellow, with green leaves interspersed among them.

Thank
You